

Demonstration Estate (legacy extraction tier)

Schema **RECON_TEST** · Oracle 21.0.0.0.0 · extracted 2026-08-21 21:50:47
Prepared by DevCrafter · 2026-08-23 · engine: pgrecon



Executive summary

This assessment covers 35 database objects and 150 lines of PL/SQL in the client's Oracle 21.0.0.0.0 environment, from which 43 discrete migration findings were identified. The overall migration grade is C, indicating a major program of work for which phased delivery is essential. The finding set spans 9 high-severity, 18 medium-severity, 14 low-severity, and 2 informational items, reflecting a codebase with a modest number of objects but a meaningful concentration of constructs that do not translate directly to PostgreSQL.

Estimated effort to complete the migration ranges from 97.8 to 165.4 person-days, with an expected case of 120.3 person-days. This range should be treated as a planning input rather than a fixed commitment, and it will narrow as remediation proceeds and the high-severity items are resolved or reclassified. Given the grade and effort spread, the client should plan for a structured, multi-stage program rather than a single conversion pass.

43 FINDINGS	9 HIGH	18 MEDIUM	120 EXPECTED PERSON-DAYS
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Effort range: **98–165** person-days (4.7–7.9 person-months), expected 120 — including testing and stabilization. Assumptions are listed with the estimate; this is a scoping instrument to be calibrated per engagement, not a quote.

Estate overview

35 OBJECTS	150 PL/SQL LINES	5/5 DDL PARSED	9 UNITS DEEP-PARSED
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OBJECT TYPE	COUNT
TABLE	10
INDEX	8
SEQUENCE	3
VIEW	2
TRIGGER	2
PROCEDURE	2
TYPE BODY	1
TYPE	1
SYNONYM	1
PACKAGE BODY	1
PACKAGE	1
MATERIALIZED VIEW	1
JOB	1
FUNCTION	1

Largest tables by rows (from optimizer statistics)

TABLE	ROWS
SALES	50
EMP	4
DEPT	3
MV_DEPT_SALARIES	3
AUDIT_LOG	0
INVOICES	0
LEGACY_NOTES	0

LEGACY_REFS	0
PRODUCT_SPECS	0

License exposure

What the database reports about its own edition, hardware, and use of separately licensed features. These are the source system's facts; contract interpretation and pricing belong in the engagement, next to the actual agreements.

Express EDITION	3 CPU CORES	2 CPU_COUNT IN FORCE
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LICENSED OPTION	EVIDENCE FEATURE	IN USE NOW	DETECTIONS	LAST USED	NOTE
Partitioning	Partitioning (user)	yes	1	2026-08-18	Enterprise Edition option

Every separately licensed option in use is cost that ends with the migration; every one in use without a matching license entry in the client's agreements is audit risk that ends with it too.

Migration hot spots

The ten objects concentrating the most remediation weight; these decide the project's shape and deserve design attention first.

OBJECT	FINDINGS	WEIGHT	DRIVEN BY
RECON_TEST.PKG_LEDGER	6	12.3	Package-level state (5), Package initialization block (2), PL/SQL collection type (2), +3 more
RECON_TEST.ARCHIVE_NOTES	8	8.3	UTL_FILE usage (3), ROWID in code (1.5), DBMS_LOB usage (1.5), +5 more
RECON_TEST.TRG_EMP_AUDIT	2	5.5	Autonomous transaction (4), Transaction control inside PL/SQL (1.5)
RECON_TEST.PKG_LEDGER (body)	1	5.0	Package-level state (5)
RECON_TEST.TRG_EMP_SAL_GUARD	2	4.3	Compound trigger (4), SYSDATE (0.3)
RECON_TEST.LEGACY_NOTES.BODY	1	4.0	LONG or LONG RAW column (4)

RECON_TEST.MV_DEPT_SALARIES	2	3.5	Materialized view (2), Query-rewrite materialized view (1.5)
RECON_TEST.LOOPBACK	1	3.0	Database link (3)
RECON_TEST.LEGACY_REFS.SCAN_DOC	1	3.0	BFILE column (3)
RECON_TEST.PRODUCT_SPECS.SPEC	1	3.0	XMLTYPE column (3)

Findings and remedies

HIGH Database link R-OBJ-01 • extension: postgres_fdw

Cross-database access needs postgres_fdw or oracle_fdw on the PostgreSQL side, plus credential management. Every statement using the link must be found and repointed.

RECON_TEST.LOOPBACK to //localhost:1521/XEPDB1 as RECON_TEST

HIGH Global index on a partitioned table R-PERF-02

PostgreSQL indexes on partitioned tables are always local: there is no global index, and a unique key must contain the partition key. Global-index lookups become per-partition probes, and unique constraints that cannot include the key force a design decision, not a translation.

RECON_TEST.SALES_AMOUNT_GIX GLOBAL partitioned index on SALES

HIGH Package-level state R-PKG-01

PostgreSQL has no package variables. Scalars map to session settings via set_config and current_setting (text only); composites and collections need per-session temporary tables. Every reader and writer of the state changes.

RECON_TEST.PKG_LEDGER	2 package-level declaration(s), first at line 2
RECON_TEST.PKG_LEDGER (body)	1 package-level declaration(s), first at line 2

HIGH UTL_FILE usage R-SYS-01 • extension: orafce

PostgreSQL functions have no supported server-side file API. Move file handling to the application, or accept the operational cost of an untrusted language such as plpython.

RECON_TEST.ARCHIVE_NOTES

UTL_FILE (first at line 16)

HIGH Compound trigger R-TRG-01

PostgreSQL has no compound triggers. Split the timing sections into separate trigger functions and move statement-level state into a transition table or a session variable.

RECON_TEST.TRG_EMP_SAL_GUARD

on EMP

HIGH Autonomous transaction R-TRG-02

PostgreSQL functions cannot commit independently of their caller. Rework audit-style autonomous transactions with a dblink loopback, a message table drained by a worker, or pg_background.

RECON_TEST.TRG_EMP_AUDIT

PRAGMA AUTONOMOUS_TRANSACTION (first at line 5)

HIGH LONG or LONG RAW column R-TYPE-01

LONG has been deprecated since Oracle 8. Map LONG to text and LONG RAW to bytea; any code reading the column piecewise must be rewritten, and only one LONG per table exists to find.

RECON_TEST.LEGACY_NOTES.BODY

LONG

HIGH BFILE column R-TYPE-07

BFILE points at a file outside the database and PostgreSQL has no counterpart. Store a path or object key plus the file in application storage, and move the DIRECTORY grants and BFILENAME plumbing into the application.

RECON_TEST.LEGACY_REFS.SCAN_DOC

BFILE

MEDIUM Scheduler or legacy job R-OBJ-02 • extension: pg_cron

DBMS_SCHEDULER and DBMS_JOB have no engine equivalent in PostgreSQL. Recreate schedules in pg_cron and port each job action; calendar syntax must be translated by hand.

RECON_TEST.NIGHTLY_ROLLUP

scheduler job

MEDIUM Materialized view R-OBJ-03

PostgreSQL materialized views refresh only in full and have no refresh-on-commit. Fast-refresh views need a trigger-based design or a scheduled REFRESH with the staleness accepted.

RECON_TEST.MV_DEPT_SALARIES

materialized view

MEDIUM User-defined object type R-OBJ-05

Composite types carry the attributes, but member functions do not exist in PostgreSQL: each method becomes a standalone function taking the composite as its first argument, and every call site changes.

RECON_TEST.T_MONEY

object type

MEDIUM Query-rewrite materialized view R-PERF-05

Oracle rewrites matching queries to hit the materialized view transparently; PostgreSQL never does. Queries that benefited must reference the view explicitly, which means finding them in the application first.

RECON_TEST.MV_DEPT_SALARIES

REWRITE enabled, refresh COMPLETE

MEDIUM Package initialization block R-PKG-02

Oracle runs the block once per session on first reference, which PostgreSQL cannot replicate. Extract it into an init function guarded by a session flag and call it lazily from each entry point. Detection is heuristic; confirm by eye.

RECON_TEST.PKG_LEDGER

BEGIN blocks exceed subprograms by 1 (possible initialization block)

MEDIUM Dynamic SQL R-SRC-01

EXECUTE IMMEDIATE maps to PL/pgSQL EXECUTE, but bind syntax and error handling differ. Each dynamic statement needs a manual port and a test.

RECON_TEST.PKG_LEDGER

EXECUTE IMMEDIATE (first at line 27)

MEDIUM GOTO statement R-SRC-02

PL/pgSQL has no GOTO. The control flow must be restructured with loops, exceptions, or early returns.

RECON_TEST.FIND_MANAGER

GOTO (first at line 14)

MEDIUM BULK COLLECT R-SRC-03

PL/pgSQL has no BULK COLLECT; array_agg into an array or a plain set-based statement usually replaces the whole loop.

RECON_TEST.PKG_LEDGER

BULK COLLECT (first at line 24)

MEDIUM CONNECT BY hierarchical query R-SRC-04

Rewrite as a recursive CTE (WITH RECURSIVE). LEVEL, SYS_CONNECT_BY_PATH, and ORDER SIBLINGS BY all need manual equivalents.

RECON_TEST.V_ORG_CHART

CONNECT BY in view definition

MEDIUM Transaction control inside PL/SQL R-SRC-14

PostgreSQL functions cannot COMMIT or ROLLBACK; only procedures called outside a transaction block can. Each routine that controls transactions must become a procedure or hand that control back to the caller.

RECON_TEST.TRG_EMP_AUDIT

COMMIT or ROLLBACK (first at line 15)

MEDIUM PL/SQL collection type R-SRC-16

Nested tables, varrays, and associative arrays map to PostgreSQL arrays, jsonb, or temporary tables depending on use; collection methods such as EXTEND, COUNT, and FIRST have no direct equivalents and every reader changes.

RECON_TEST.PKG_LEDGER

collection type declaration (first at line 20)

MEDIUM Empty string treated as NULL R-SRC-18

Oracle treats "" as NULL; PostgreSQL treats it as a real value. `v := ''` assigns NULL in Oracle but an empty string in PostgreSQL, `NVL(x, '')` stops collapsing, and `col = ''` can start matching rows. Nothing errors: behavior changes silently. Decide per site whether NULL or empty was meant, and put the data path under test.

RECON_TEST.ARCHIVE_NOTES

empty-string literal (first at line 4)

MEDIUM ROWID in code R-SRC-19

ctid is not a ROWID: it moves on update and vacuum and must not be stored or compared. Fetch-by-rowid loops rewrite to primary-key access or WHERE CURRENT OF; ROWID-typed variables and parameters change type with them.

RECON_TEST.ARCHIVE_NOTES

ROWID (first at line 3)

MEDIUM DBMS_LOB usage R-SYS-04

Piecewise LOB calls mostly disappear: text and bytea are handled whole, with substring and length built in. Streaming access beyond 1 GB needs large objects and the `lo_` API.

RECON_TEST.ARCHIVE_NOTES

DBMS_LOB (first at line 14)

MEDIUM Global temporary table R-TAB-01 • extension: pgtt

PostgreSQL temporary tables are per session and vanish with it; there is no shared definition with private rows. Rework to on-the-fly temp tables or use the pgtt extension.

RECON_TEST.STAGING_ROWS

GLOBAL TEMPORARY

MEDIUM XMLTYPE column R-TYPE-02

PostgreSQL xml stores documents but has no XML index type and none of the XMLType member functions; XQuery support differs. Inventory how the application queries the XML before mapping.

RECON_TEST.PRODUCT_SPECS.SPEC

XMLTYPE

MEDIUM TIMESTAMP WITH LOCAL TIME ZONE column R-TYPE-03

Values are normalized to the session time zone on read, which timestamptz does not replicate exactly. Verify every reader assumes UTC or one fixed zone before mapping to timestamptz.

RECON_TEST.INVOICES.CREATED_AT

TIMESTAMP(6) WITH LOCAL TIME ZONE

MEDIUM ROWID or UROWID column R-TYPE-06

A stored ROWID is a physical row address, and PostgreSQL has no stable one: ctid changes on update and vacuum. Replace the column with a real key reference; whatever wrote and followed these addresses changes with it.

RECON_TEST.LEGACY_REFS.REF_ADDR

ROWID

LOW Synonym R-OBJ-06

PostgreSQL has no synonyms. Same-database synonyms usually dissolve into search_path; cross-schema renames become views or schema qualification at the call sites.

RECON_TEST.STAFF

for RECON_TEST.EMP

LOW Partitioned table R-PART-02

Declarative partitioning covers RANGE, LIST, and HASH, but global indexes and reference partitioning do not exist. Each partitioned table needs its scheme mapped explicitly.

RECON_TEST.SALES

RANGE

LOW **Optimizer hint** R-PERF-01 • extension: pg_hint_plan

PostgreSQL ignores optimizer hints; pg_hint_plan can pin plans but is a last resort. A hinted statement was tuned by hand against Oracle's planner, so budget a fresh look at its plan and indexes on PostgreSQL instead of a plain rewrite.

RECON_TEST.ARCHIVE_NOTES

optimizer hint (first at line 8)

LOW **Parallel degree set on object** R-PERF-03

PostgreSQL has no per-object parallel degree; the planner decides within max_parallel_workers_per_gather. Workloads that relied on forced parallel scans need a plan review and possibly session-level parallel settings.

RECON_TEST.SALES

DEGREE 4

LOW **Old-style outer join (+)** R-SRC-05

The (+) operator does not exist in PostgreSQL. Rewrite as ANSI LEFT or RIGHT JOIN; semantics match except for a few multi-condition corner cases worth testing.

RECON_TEST.V_EMP_DEPT

uses (+) outer join

LOW **SYSDATE** R-SRC-11

SYSDATE is evaluated at call time; now() is fixed at transaction start. Map to statement_timestamp() or clock_timestamp() where the difference matters, for example inside long transactions and loops.

RECON_TEST.ARCHIVE_NOTES

SYSDATE (first at line 19)

RECON_TEST.PKG_LEDGER

SYSDATE (first at line 7)

RECON_TEST.TRG_EMP_SAL_GUARD

SYSDATE (first at line 8)

LOW **DBMS_OUTPUT usage** R-SYS-05 • extension: orafce

Map PUT_LINE to RAISE NOTICE for diagnostics. If callers consume the output programmatically, that pattern needs a real interface instead.

RECON_TEST.ARCHIVE_NOTES

DBMS_OUTPUT (first at line 19)

LOW NUMBER without precision R-TYPE-04

Bare NUMBER accepts up to 38 significant digits. Map to numeric, and check whether application code assumes integer behavior that a typed bigint would express better.

RECON_TEST.AUDIT_LOG.AUDIT_ID	NUMBER with no precision
RECON_TEST.INVOICES.INVOICE_ID	NUMBER with no precision
RECON_TEST.MV_DEPT_SALARIES.HEADCOUNT	NUMBER with no precision
RECON_TEST.MV_DEPT_SALARIES.TOTAL_SAL	NUMBER with no precision
RECON_TEST.V_ORG_CHART.DEPTH	NUMBER with no precision

INFO Swallowed exception R-SRC-07

WHEN OTHERS THEN NULL hides every failure including the ones a migration introduces. Worth removing before the port so the test phase can actually see errors.

RECON_TEST.ARCHIVE_NOTES	WHEN OTHERS THEN NULL (first at line 22)
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INFO Table without optimizer statistics R-STAT-01

num_rows is null, so data-volume estimates for this table are unreliable. Gather statistics before trusting sizing numbers.

RECON_TEST.STAGING_ROWS	no statistics gathered
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Risk outlook

The high-severity findings, though limited to nine occurrences, touch on constructs that carry outsized migration risk if not addressed early: a database link, a global index on a partitioned table, package-level state used in two places, UTL_FILE usage, a compound trigger, an autonomous transaction, a LONG or LONG RAW column, and a BFILE column. Each of these represents a pattern with no direct PostgreSQL equivalent or one that requires materially different handling, and their presence concentrated in a small number of objects means that targeted remediation of those objects will disproportionately reduce overall risk. The medium-severity findings, numbering eighteen, are more varied - covering scheduler jobs, materialized views, dynamic SQL, object types, and similar constructs - and while individually less severe, their volume adds to the overall conversion effort. The low and informational findings are largely items of code hygiene and precision that should be addressed but do not threaten the migration timeline.

Recommended approach

Given the severity mix, with high-severity issues outnumbered by medium-severity ones but carrying greater individual risk, the recommended approach is to phase the work by addressing the high-severity constructs first, since they are concentrated and their resolution will de-risk the remainder of the program. Analysis of the hotspots shows that findings are not evenly distributed across the 35 objects: a small number of objects account for a disproportionate share of both finding count and weighted complexity, and prioritizing remediation on these objects first will yield the greatest reduction in overall risk per unit of effort. Subsequent phases can then work through the broader set of medium-severity findings before closing out the low-severity and informational items, allowing the client to sequence the program in line with the effort range estimated above.

Effort estimate

COMPONENT	PERSON-DAYS
baseline and environment	5.0
schema conversion	1.8
finding remediation	67.6
PL/SQL porting by volume	0.8
data movement	0.0
Development subtotal	75.2

With testing and stabilization: **98 / 120 / 165** person-days (low / expected / high).

Assumptions

- calibration profile 'default': the public engine's conservative defaults, unchanged
- stored code ports at 200 lines per person-day including unit tests
- repeated findings of one rule cost a severity-dependent fraction of the first fix
- testing and stabilization runs 1.3x to 2.2x development, 1.6x expected
- static estimate from schema facts alone: workload, data quality, and application coupling are not visible here

Appendix: findings detail

Each rule lists up to twelve objects here so repeated findings stay readable at estate scale; the findings CSV delivered with this report carries every object with its evidence.

RULE	SEVERITY	OBJECT	EVIDENCE
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R-OBJ-01	high	RECON_TEST.LOOPBACK	to //localhost:1521/XEPDB1 as RECON_TEST
R-PERF-02	high	RECON_TEST.SALES_AMOUNT_GIX	GLOBAL partitioned index on SALES
R-PKG-01	high	RECON_TEST.PKG_LEDGER	2 package-level declaration(s), first at line 2
R-PKG-01	high	RECON_TEST.PKG_LEDGER (body)	1 package-level declaration(s), first at line 2
R-SYS-01	high	RECON_TEST.ARCHIVE_NOTES	UTL_FILE (first at line 16)
R-TRG-01	high	RECON_TEST.TRG_EMP_SAL_GUARD	on EMP
R-TRG-02	high	RECON_TEST.TRG_EMP_AUDIT	PRAGMA AUTONOMOUS_TRANSACTION (first at line 5)
R-TYPE-01	high	RECON_TEST.LEGACY_NOTES.BODY	LONG
R-TYPE-07	high	RECON_TEST.LEGACY_REFS.SCAN_DOC	BFILE
R-OBJ-02	medium	RECON_TEST.NIGHTLY_ROLLUP	scheduler job
R-OBJ-03	medium	RECON_TEST.MV_DEPT_SALARIES	materialized view
R-OBJ-05	medium	RECON_TEST.T_MONEY	object type
R-PERF-05	medium	RECON_TEST.MV_DEPT_SALARIES	REWRITE enabled, refresh COMPLETE
R-PKG-02	medium	RECON_TEST.PKG_LEDGER	BEGIN blocks exceed subprograms by 1 (possible initialization block)
R-SRC-01	medium	RECON_TEST.PKG_LEDGER	EXECUTE IMMEDIATE (first at line 27)
R-SRC-02	medium	RECON_TEST.FIND_MANAGER	GOTO (first at line 14)
R-SRC-03	medium	RECON_TEST.PKG_LEDGER	BULK COLLECT (first at line 24)
R-SRC-04	medium	RECON_TEST.V_ORG_CHART	CONNECT BY in view definition
R-SRC-14	medium	RECON_TEST.TRG_EMP_AUDIT	COMMIT or ROLLBACK (first at line 15)
R-SRC-16	medium	RECON_TEST.PKG_LEDGER	collection type declaration (first at line 20)
R-SRC-18	medium	RECON_TEST.ARCHIVE_NOTES	empty-string literal (first at line 4)
R-SRC-19	medium	RECON_TEST.ARCHIVE_NOTES	ROWID (first at line 3)
R-SYS-04	medium	RECON_TEST.ARCHIVE_NOTES	DBMS_LOB (first at line 14)
R-TAB-01	medium	RECON_TEST.STAGING_ROWS	GLOBAL TEMPORARY

R-TYPE-02	medium	RECON_TEST.PRODUCT_SPECS.SPEC	XMLTYPE
R-TYPE-03	medium	RECON_TEST.INVOICES.CREATED_AT	TIMESTAMP(6) WITH LOCAL TIME ZONE
R-TYPE-06	medium	RECON_TEST.LEGACY_REFS.REF_ADDR	ROWID
R-OBJ-06	low	RECON_TEST.STAFF	for RECON_TEST.EMP
R-PART-02	low	RECON_TEST.SALES	RANGE
R-PERF-01	low	RECON_TEST.ARCHIVE_NOTES	optimizer hint (first at line 8)
R-PERF-03	low	RECON_TEST.SALES	DEGREE 4
R-SRC-05	low	RECON_TEST.V_EMP_DEPT	uses (+) outer join
R-SRC-11	low	RECON_TEST.ARCHIVE_NOTES	SYSDATE (first at line 19)
R-SRC-11	low	RECON_TEST.PKG_LEDGER	SYSDATE (first at line 7)
R-SRC-11	low	RECON_TEST.TRG_EMP_SAL_GUARD	SYSDATE (first at line 8)
R-SYS-05	low	RECON_TEST.ARCHIVE_NOTES	DBMS_OUTPUT (first at line 19)
R-TYPE-04	low	RECON_TEST.AUDIT_LOG.AUDIT_ID	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.INVOICES.INVOICE_ID	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.MV_DEPT_SALARIES.HEADCOUNT	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.MV_DEPT_SALARIES.TOTAL_SAL	NUMBER with no precision
R-TYPE-04	low	RECON_TEST.V_ORG_CHART.DEPTH	NUMBER with no precision
R-SRC-07	info	RECON_TEST.ARCHIVE_NOTES	WHEN OTHERS THEN NULL (first at line 22)
R-STAT-01	info	RECON_TEST.STAGING_ROWS	no statistics gathered

Produced with the pgrecon assessment engine from an offline extraction; no connection was made to the source database. Findings are deterministic and reproducible from the same inventory. DevCrafter · devcrafterai.com